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Summary

Java Backend Engineer with applied LLM and platform-adjacent project breadth and 4.6+ years delivering enterprise systems across **healthcare payer integrations** and **automotive configuration and recommendation products**.

Delivery scope: Carries backend capabilities from requirements and technical analysis through Java/Spring implementation, API and database integration, testing, CI/CD validation, observability, incident analysis, and release support; extends this foundation through Python/FastAPI, LLM evaluation, React, and AWS event-driven projects.

Engineering approach: Builds reliable integrations and reusable operational capabilities by tracing behavior across service, SQL, messaging, and downstream boundaries, preserving deterministic validation, and measuring latency, failures, volume, and model quality.

Working style: Collaborates across engineering, QA, database, deployment, vendor, and client-facing teams; documents repeatable solutions, reviews code, mentors developers, and uses production and evaluation evidence to guide improvements.

Technical Skills

Backend & APIs: Java 8/11, Core Java, Spring Boot, Spring Framework/MVC, REST/SOAP APIs, JAX-WS, Java EE/J2EE, EJB, Hibernate/HQL, JPA, JDBC/OJDBC, Python, FastAPI, OOP, Design Patterns.

AI/LLM Project Technologies: PyTorch, Hugging Face Transformers, T5, PEFT/LoRA, adapter lifecycle, summarization and classification, offline ROUGE evaluation, model diagnostics, Prometheus.

Messaging, Data & Cloud: JMS, Azure Service Bus/Relay/Functions; SQL, Oracle, DB2, PostgreSQL; AWS S3, Lambda, EFS, ECS, RDS, CloudWatch, IAM, VPC; Docker, Terraform; Kafka (project), Kubernetes (knowledge).

Delivery & Observability: Jenkins, Maven, GitLab CI/CD, GitHub Actions, IBM UrbanCode, Git, SVN; JUnit/JUnit4, Mockito, Integration Testing, SoapUI, Postman; Log4j2, correlation IDs, latency, success/failure, and request-volume telemetry.

Frontend & Practices: React, TypeScript, Vite, Tailwind CSS (projects); Agile/Scrum, Root-Cause Analysis, Performance Tuning, Legacy Modernization, Technical Documentation, Code Reviews, Mentoring.

Selected Projects

Privacy-Preserving AI Email Client (Graduate Capstone)

- Built an **LLM-powered email summarization and classification system** with a Dockerized FastAPI model service, PyTorch, Hugging Face Transformers, T5, and PEFT/LoRA, plus MinIO, Prometheus, React, and TypeScript.
- Implemented model configuration, summarization, categorization, adapter lifecycle, and diagnostic workflows, exposing AI capabilities through service APIs and a web client.
- Developed an **offline evaluation workflow** and measured the fine-tuned model at **ROUGE-1 F 53.04, ROUGE-2 F 44.90, and ROUGE-L F 45.90**, using repeatable quality metrics to assess model output.

AWS Automated Spark Pipeline

github.com/preferablehuman/AWS-automated-spark-pipeline

- Built an **event-driven AWS data workflow** where S3 notifications invoke Lambda staging to EFS, Spark Structured Streaming processes micro-batches in an ECS-hosted Docker container, and results persist to RDS PostgreSQL over JDBC.
- Provisioned VPC, IAM, S3, Lambda, EFS, ECS, RDS, security groups, and CloudWatch resources with Terraform; processed an **8GB+ dataset** and persisted checkpoints for restart recovery and operational visibility.

Work Experience

NTT Data, Bengaluru, India

Software Developer, Aug 2022 - Aug 2024

- Developed and maintained **Java 8/11, JAX-WS/Spring, and REST/SOAP** healthcare integration services on IBM WebSphere, supporting **50K+ daily requests at 99.99% SLA availability**.
- Integrated hybrid service flows across **Azure Relay, Azure Service Bus, Azure Functions, and JMS**, synchronous downstream REST calls, and Oracle/OJDBC processing with explicit timeout and partial-failure handling.
- Integrated a reusable AOP-based monitoring library that propagated correlation IDs and captured **latency, success/failure, and request-volume telemetry**, publishing operational data to a central monitoring application.
- **Led a project-wide Log4j2 migration**, owning the initial PoC, preserving production logging contracts, documenting a repeatable upgrade path, and reviewing the team rollout.
- Supported root-cause analysis of a WebSphere memory/resource-retention incident through code-path review, mitigation planning, and rolling-restart validation while maintaining service availability.
- Supported Jenkins/Maven/IBM UrbanCode workflows, JUnit/SoapUI validation, post-release checks, production incident analysis, and go/no-go or rollback recommendations.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Developed and supported a Java EE/J2EE, JSF, Hibernate/HQL enterprise application on Oracle WebLogic with DB2 for Mercedes/Daimler Vans configuration, recommendation, and ordering workflows.
- Integrated an asynchronous Python vehicle-configuration prediction workflow with Java services and callbacks, then enforced deterministic option and package validation before recommendations reached dealer users.
- **Reduced VIN repeat-configuration API latency from 60+ seconds to 10–20 seconds** by moving stable option-code mapping into set-based HQL/DB2 expressions while retaining business validation in Java.
- Diagnosed cross-system issues across Python output, Java validation logs, and DB2 reference data; supported Jenkins/GitLab CI/CD, JUnit4, code reviews, and production debugging.
- **Mentored junior developers** through codebase walkthroughs, incremental defects and features, debugging support, and code reviews.

Capgemini, Mumbai, India

Analyst Trainee, Jan 2020 - Jun 2020

- Built Java Spring/Spring MVC, Angular, and PostgreSQL modules for fitness-club and digital-wallet applications, implementing authentication, email/OTP login, CRUD, balances, transactions, and profile workflows.

Education

Central Michigan University, Mount Pleasant, MI, USA

Aug 2024 - Dec 2025

Master of Science in Computer Science, GPA: 3.87/4.0

Lovely Professional University, Punjab, India

Aug 2016 - May 2020

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

Certifications

Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate